ActInf Textbook Group ~ Cohort 8 ~ Session 1 (Onboarding and Chapter 1) ~ 2/20/2025

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all right hello welcome cohort 8 um thank you all for sharing your um hellos in the first minutes so for the rest of this meeting briefly we'll cover the meeting structure for this interval of activity and a little bit about our Niche and uh to whatever extent you'll come to explore it and modify it as well and then um we'll kind of discuss one foot in chapter one one foot in any other general questions because that's kind of where chapter 1 and 10 are at they're like book ends they kind of lead in and lead out so it is a good approach chapter as behooves of chapter one so this uh cohort 8 live meeting is our landing page if you want to bookmark that one it's a good one um and as briefly mentioned cohort 7 will be active at the same time weekly going through the second half of the book so you're welcome to join you're welcome to rewatch videos from any cohort you're welcome to join live at the same time slot weekly at which um time alternating weeks it'll be from the second half of the book like next week at this time will'll be people saying hello in cohort 7 slh whoever is there and going over chapter 6 and then when we're on chapter 2 they'll be on chapter 7 and so on um and and the the the day of the meeting a recording is posted so it's kind of like the synchronous element and the rhythm of two weeks per chapter um with uh lots of edges coming in and out so to briefly go over this Koda document um at any time anyone can write a question in the zoom chat or raise your hand in Zoom so just any anything that you want to clarify or or um just go for it um and uh it this document has like open comment access and then also you can request edit access if for some reason like your email just hasn't been added to to edit the the to add directly so um top page you can right click and download the PDF of the textbook and it's just a welcome page directory has some pages linked um onboarding this this is like some more information about onboarding to The Institute overall there's a lot of other projects and activities outside of the textbook group so check it out and also learning practices like we come from many different backgrounds we have many different time availabilities many different um affordances are are accessible to us we have many different preferences for where we go and so on so um adding different activities that is like okay when we time box and um you know C in to be learning and applying active inference what is that really looking like and uh what's what's a a healthy and a productive blend for us and uh some of those are solitary some of those are in your own Knowledge Management environment some of those might be in this open source knowledge environment or in another repository somewhere um contribute to the textbook group here again there's more ways to do than not to do with contributing so if you feel like yeah I'd love to improve the synchronous or the asynchronous components of this group um or or there's other places to contribute outside of textbook group too so that Journey might take you there but it could be anything from playing at some point like a live role in the meetings for example the later time slot uh is going to be facilitated by Andrew Pasa who has joined textbook group also so many times and and will will pitch fast balls also and facilitate those discussions um taking notes and and describing developing different code examples um asking questions this is going to be like a totally vital element like a lot of times it can feel like what we don't understand is you know our limitation or something but actually when we externalize that question we can make a lot of interesting learnings and progress together so check out different ways to contribute if that is exciting the meetings page has the past completed cohorts and the two ongoing cohorts so that's kind of like the like organizational about the textbook group and you can fold pages and Koda like that and also you can make it light or dark if if it looks different on your screen but I'm using the dark version all right then there's two other main sections there's the textbook and then there's kind of our discourse so first the textbook itself we have the pages in a table and you can download these tables or you know make but these it may become relevant and also Koda is really cool can make your own Koda or you can play around but by putting at you can tag any page or person or even row so like at page one a mouse over shows that or at surprise is going to be a link to the active inference ontology or different other places where it's been mentioned thank you for posting Nick awesome so that's the Page by Page and also you can click on someone's uh token icon and you'll jump to their location in Kota so if you click on the 4D you'll you'll jump to where I am um so it's just Page by Page okay uh then we get to the chapters so there's 10 chapters and three appendices um there's detailed table of contents just skimming and just familiarizing and many Coates of paint will just come up again and again with just like kind of taking a deep breath and just reading it and just you know there's stuff that's erroneous that we've discovered there's stuff that is ambiguous or or over under precise or just there's there's so many different readings but just sticking with it on a first read and staying in the game you'll be reducing your uncertainty then there's the sections here so this is kind of where we go when we're going to have like the chapter one discussion and there's there's there's a few main each of these chapter Pages have a few main components first there's a um kind of like a 20 minutu is overview that Oli and I did over several days these are um you know they're fun um and we just sort of contextualize like what is this chapter doing in the context of the book and just what are the what are the sections of the chapter basically then there's these questions that are all part of this underlying questions table which we're going to get back to but you can upvote questions that you think are interesting and we'll kind of always prioritize or frontload written questions that people have added in and then when um and then live questions that people have that will enter into arose and then you know failing that we kind of just return to highly interesting questions and then this canvas has some questions are answerable debatable others are more just discurs so you can always check out that canvas and read and add more you know no need to delete anything per se um but add more and and um so it goes so that's like the questions section of each page and then at the bottom there's uh sometimes short sometimes vast a little bit more unstructured so usually when we're having a discussion there if it's like kind of outside of the um realm of a question per se sometimes we're just like or just it's like a question but we just are doing out in the open it's not a fully formed question yet um we we will um put it there so it's like everything's in a state of Open Source and contributive enjoyment and you know does active inference currently allow for learning a model structure it's like it does but that's not reflected here but if you see that and you go yeah and I'm curious about this then you know add it there or you can select text and add a comment or you could add a link that's relevant and so everything is just if you see some discrepancy between like what you'd want that page to be and what it is in your observation you can change your mind or you can change the page basically as we will come to see so that's how the chapters are overview video questions and disc course for everybody's um addition and and uh reading and then some more scattered topics over the years that people have discussed so that's the chapters then there's figures so here it's like screenshots and just the figure names and this comes up all the time because we'll tag like at figure 1.1 and then you can get a mouse over and you can see these figures and and these are also helpful just to to see and to kind of Be Inspired technically or to to make some artistic rendering of or all this um the equations section so here are screenshots similarly of the taggable equations like you can just say like equation 2.1 then um for those who want to explore a little bit of latch and and uh using Ai and so on here's latch re renderings of the equations and in this latch column is the plain text latch so if you want to you can utilize those and then using just the simple Koda generative AI from the latek here's like explanation summary code where is it important what are the variables now these are just sort of a little bit like a low context Generation Um about what what are some related but but their conversation starters for sure to just go from the latch of Base equation these are some you know and then do you have it you know would you phrase one differently would you add one would you remove one it's not relevant like that's already you're in the sort of metacognitive and learning and relevance realization process by exploring it that way whatever your level of math familiarity is okay thank you Marvin I'll will return to that um boxes and tables these are just similar for the equations and figures they're just for the boxes and the tables so that's just this section is just the um kind of parsing of the textbook itself I'm just going to return briefly to to that um important piece I didn't uh mention so this is like when everybody registered they gave a a self assess 0 to 100 on four questions which is how familiar are you with active inference and then also how familiar are you with Computer Science Biology and math which are kind of like three of the disciplinary areas that these conversations are are related to so the xois is familiarity with active inference so for example person who put 100 let's just say um they also rated themsel as 78 or whatever in biology and 100 in math so what this is showing there's a few interesting patterns here first intering pattern is there's a sharp psychological drop off of people at 80 very few people rate themselves as familiar with active Ence above 80 again this doesn't even this is all self assessed so so um you you could um you could even you know smile and chuckle at some of these but there's and then there's and then it doesn't show the density but there's a strong at 50 like maximal uncertainty and then like other than that what this shows is we run the self assessed and probably externally assessed gamut from those with little familiarity with math computer science and biology and Neuroscience to very very high expertise in those areas so that's just kind of the situation and the fun is we we can't necessarily presuppose that any given um technique or topic is going to be familiar to others like just we have we're we're we're living in different places in the world and we have different educational backgrounds and all of that but this kind of just in one visualization shows that a little bit with a like almost complete pain painting of how familiar we are with these different topics okay and then just this last um main section discourse so here's a lot of the things that people have added over the years so again questions are a key aspect and you can add questions by by um going to the bottom of some uh subsection like here we're in chapter one and you just can go to the bottom I can you can make a button too to add it and then just hitting enter there and then you're adding a question and that will show up in the chapter one filtered view so or or there's you know so there's ways to add questions um applying active inference this one is is going to come come up again and again I mean learning and applying active inference is kind of the bigger picture that's what we do at The Institute so if that is your interest um you also may want to check out like there's there's many many links and we can return to this more specifically we're not going to jump in at this minute to just where are all the applications hosted and all that but they're out there um and there's more to make um math learning group is kind of like a special subgroup um that has done different things over the seasons last uh cohort activity uh Dr Octopus was leading like a group and so there's some sessions with octopus and and he did a lot of other work too there's math learning resources so these are some background information but like everyone's going to be in a different place and this is going to trickle out into learning like probability and stochastic systems dynamical systems linear algebra all these different topics that are explored a little bit in the appendix they're just kind of noted um and also nowadays using chat-based generative AI is an excellent way to learn and go back to those um foundations um also the SPM textbook is a great place to go for understanding friston at all's trajectory into active inference um notation we can return to again later with equations and stuff there's a lot there math learning questions these are kind of like meta questions about math learning um math overviews some um people have written and contributed just to to more uh just to breakdowns of the chapters about certain math things so but again this just shows like there's there's people who want to um you know for whom it will be the first time seeing a basian equation or for whom linear regression or matrix multiplication may be unfamiliar and then also there are those who want to really explore the the analytical rigor and understand bayy mechanics and all these other topics at a really advanced level and uh you're either if if you leave a trace you're either going to be contributing to important educational materials that will help others immensely Andor contributing to Frontier research so go for it um projects this is a table where people can like list kind of concepts for projects some of these are actually did very much continue on more broadly the activities page activities. active inference do Institute will link you to like ongoing projects at The Institute other things that are hosted here and then other projects that are happening in the ecosystem so if you want to just get involved in projects that have little to do with the textbook there's a lot of ways that you can find out about that and propose your own and then here is um just like things that people have ideas people have brought up related to the textbook group itself um code we there's we we'll return to it but yeah SPM textbook that's statistical parametric mapping so important little historical Trivium okay um the reason why Carl friston is the most cited neuroscientist even sort of outside SL prior to the free energy principle and active inference development thrust of the last like 19 is years is this statistical toolbox for processing neuroimaging data like fmri and EEG so the textbook and the documentation the textbooks last hard cover publication was 2007 and then since then there's documentation online and there's also like a recent 2025 release of the software package it's in mat lab some of the methods are written in it we we may or may not explore those this is really really good for showing in a well-worked and well-studied system of Interest like how do you go from the data that comes off of the fmri machine on through the kinds of smoothing and statistical assessments that lead to your ability to do like dynamic causal modeling and counterfactuals on base graphs and understanding and modeling cognitive behavioral systems so it's um kind of the the precursor to active inference but it it also really helps ground it in experimental paradigms um then T in terms of frisen all trajectory into active inference um I mean in terms of their historical development of studying perception and action systems empirically leading into the generalizations that are seen in active inference and free energy principle but also in terms of one of Us's individual learning trajectories because the SPM documentation textbook is a really good um if dense review of parametric basian and non-parametric statistics in a specific empirical Paradigm so that can really help ground a lot of the discussions of like statistics and probability that come up um code here's a table that um links to other documents but here's several dozen in different computer languages and there's other ones that that aren't added but I mean this is again like a classic collaborative annotation task it's like adding these different examples building them in different languages so especially in mat lab Python and Julia but there's also some other stuff in in go and some other things we can discuss um ideas and insights if it's not a question but somebody just wants to like be like yeah wow that was kind of like a really insightful conclusion of this part of the book or here's what this made me realize um resources here uh is great place to post resources Ali created this table um with a nice sorting from less technical to more technical so this is another Learning Journey to explore videos of course you can check our Channel I think earlier somebody mentioned machine learning Street talk and there's other channels that variously make um relevant material um and but also like throughout the discourse there's a lot of resources posted future textbook groups there's a um Anonymous feedback form there's also like many other many many many discussions about um ranging from um hopeful and as irational to frustrated uh about you know how do we learn and apply active inference in in this time from different backgrounds and with the tools that we have so if that's an interesting question to you there's a lot of food for thought here erata has several identified like typos or small errors some of them have been like subtly updated in um kind of sub versions in the um at the MIT press website uh others we we have sent and they may be included like just in a future kind of typo fixing edit of the textbook um so that's just some random stuff and then last section just acting tables again we'll dig more into these more topically as they come up but there's the active inference onology this is kind of like our language agnostic dialect it's that we use to interface between different natural languages and some of the technical and analytical formalisms that are used in active inference there's um this is from an earlier literature meta analysis project that some people may want to um pick back up just several hundred papers you could also just do literature searches and things like that and then also there's the um table of several hundred videos that have been produced from different series um so scanning through the papers and the um titles of guest streams even without viewing the videos can help you just get a good first pass on what the discourse is up to what are some key topics and you could search for not that the title will find everything but you know you may just s oh organizations or something like that and then you might find some relevant uh title but then also those words and there's the journal for searching more densely so that's kind of our space the synchronous meetings of course are awesome and fun and we get to see each other and have emergent discussion and everything and um also there's the time between the meetings when we engage in the learning practices you can add your information if you if you're okay with it being public because it is um you can add information to this table you can message and connect with people um in in the zoom chat but in terms of the Rhythm and just to kind of close this section we basically meet at these times in this link where we are right now and we try to keep one foot connected to the focal topic or it's sort of our strange attractor so starting with a quote or kind of stapling it to a concept that's mentioned in the text on a specific page is a great way to to begin um you know General lifelong floating uncertainties and and wonderings also can be welcome um it helps us have a convergence and uh relevant Focus to just be like okay we're in Chapter 2 you know where's a quote and then connect that and uh there's no end to the amount of pre-play you can do with preparing extensive questions and so on so of course we can't and won't linearly fit every single uncertainty and everything we have into these one hour meetings but uh between people learning and Publishing on their own blogs and repos and linking in week by week and reviewing what questions people have have thoughtfully prepared and bring to the table like it it seems far away in one sense to look at May 2025 but we can also learn a ton over these coming months so any like logistical or Niche questions otherwise we will go to chapter one and just like pick up with any page or quote or question that anyone has could you just give me an overview of the mission of The Institute that you are leading yeah to make active inference accessible and applicable and just and you're just a nonprofit doing that or what's your connection to other organizations we're Standalone volunteer nonprofit We have relationships with some other um Partners do you do you have any connections to the Santa Fe Institute not organizationally okay but if they want to they can send emails and I know that they're exploring that and and there's there's there's diffusion of peoples and ideas which is all awesome but and you know whatever streamlined and useful let's um see what happens okay um sorry I have a very concrete question uh like H how how will this uh work moving forward um between meetings we each read like the chapter and prepare as much as we want and then do we step through the chapter during the meeting or like how how will How would how would it look structurally these meetings good question like you could review previous recordings but in the end it's like who's there in the moment and then there's going to be some blend between um someone who's like um you know this is what my dissertation research is about and here's how I prepared these few things and I see these three paths but I'm not sure what you know it's like someone might have prepared a ton other people will just be like okay I I did look at it or I didn't look at chapter but I'm just here to see what's up and then um usually when it's just outside of this onboarding type session um it's basically as will go now it's like I'll open up okay we're on chapter one we can did anyone so did anyone add a written question like is there any question that somebody here added that we that's likely going to be down on the bottom that we can look at or or is there some other question that somebody here wants to look at regardless of whether it was written prior um or is there some specific quote or figure or equation so it's like for who's here is there a question or a section of the book that is um exciting for them that that that they want to go into um and and then if if again to toally failing that we can just pick a random question that either does or doesn't have discourse or just like okay Chance favors the prepared mind like what does this mean to us in the context of active inference um okay Nick is each chapter spread out over the two live meetings yeah each um each chapter occurs for two weeks but again it's like next week it'll be in that in you know seven or however many eight hours later so if you want to join then discuss chapter one again great if you want to join next week at 15 UTC it'll be discussing chapter 6 that would also be welcome but then in two weeks it'll be at this time with chapter two Sylvia I can't add a question to the table does it require edit access yeah it does require edit access just request access and I'll I'll add it when I'm just logged in differently but this new row button at the bottom if you have edit access you can just add a new row just no question oh yeah huh how do we access that panel when we're not in a zoom meeting where's the link for this um so I'm just going to put this is the cohort like this page is the one I'm looking at now you can open that in your browser anytime okay and then you'll you'll request edit access if you don't have it but everybody has comment access even like when they're not logged in um and then from there you just like chapter one that's a hyperlink to go to this chapter one page or you can navigate in the sidebar okay and then the the discourse is in this panel and you know just add add if you think it's cool um but yeah well just so you know over the next week or two people will get their their footings request edit access and I'll um provide it and then um yeah like everyone reading this will have questions the the questions will will run the gamut from like X ential philosophical to just like syntactic and semantic like what does surprise mean here and all of those spectrum of questions are really useful and identifying your uncertainty and then pursuing it from there is a huge way to learn and the more specific that that uncertainty comes to be like how did they get from equation this to that or why is this arrow in this figure like that or does it have to be then you'll come to some satisfaction and then you'll see exactly how you're learning and reducing your uncertainty through action whereas you know not writing it down or closing the book then you're not going to reduce that uncertainty you're just going to not pay attention to it but yeah what's a what's a related to chapter one or just in overview what are what are things people are are like thinking about or wondering I could have a go a question um on um on page nine on the in the first bullet point on page nine um it's talking about action and perception but then it's also saying learning is yet another way to minimize free energy however it is not fundamentally different from perception it simply operates at a slower time scale so I guess my question I have like no Neuroscience biology background like is there is there kind of an obvious sort of Intuition or something differentiating learning with perception why why they operate on different time scales I'd like to build on that question somebody gave me an interesting definition of persuasion where they claimed persuasion is just a different form of learning yeah belief update in and uh you know if someone says well the you know Carbon has a mass of 12 units and somebody's like they don't know you know they have a broad prior belief and then that updates it or whether they're being persuaded that it's 12 like that's some contextual what I I could add more thought or what does anyone think about how learn learning and perception are related I could say um for me uh for me the key

Insight of active inference is that when an organism relates with the environment there's this Freedom between two options that you can update the environment or you can update your beliefs whichever is easier and so free energy minimization uh determines which is going to be easier but so perception would be updating your beliefs in the way that learning is updating your beliefs it's just that perception is maybe one way to think about is maybe the time scale is different but another way is just the the apparatus is different but that's but they're both about updating beliefs as opposed to updating the environment thank you Andreas yeah like let's just say a ball is moving across our visual fields we could think of it as perception of the ball moving across the field or we could think about as learning and belief updating of the ball moving across the field and so learning is sometimes used to refer to slower updating processes um in a neural network like artificial neural network um perception might refer to just the the single pass forward inference whereas learning would be more about the weight updating and then even slower kinds of like structural learning could happen when the structure of the graph is changing but again like a ball moving even that is more generically described as belief updating and then whether we associate that with the phenomenology of perception and like that that it entails a perceiver and all of that or whether we think of that as doing learning on the position of the ball but then we might say like well um learning would be something that's more generic or slower time scale than perception like if we're learning the um gravitational constants we say well we perceived a bunch of balls flying across the visual field and then we did learning on some variable like what is the gravitational constants but considered from those two sides of the coin the two ways to uh reduce free energy change the mind which could be change perceptual like beliefs or change more cognitive beliefs which is associated more with learning but they're all part of the belief updating structure as opposed to reducing free energy through action but both yeah I just want to build on and ask a a question about discontinuous learning think of that in terms of Paradigm shifts or phase transitions in your belief system you know there there become moments where you know you have maybe a dramatic Insight Aha and it completely turns your mind in a New Direction so that's a different kind of learning than just sort of this continuous ball in the air I'm more interested in the in the Paradigm Shift learning is when you watch a movie for example there is learning that you get in every scene about a character about what's really going on but every now and then you get a real turn in the movie a real pivot in the movie where it shifts your mind in a totally new Direction and it's the phase transitions that really really I think uh change the search function for the engine I guess because it now looks a little bit more like a a random walk or something yeah I mean whether that is um belief updating in a discrete belief space which is discontinuous whether it's a extremely rapid continuous update that when viewed out looks like a step function whether it's structure learning there's a lot of ways to to Nuance that and even to connect it to Paradigm shifting and and discontinuous change like that um Andreas and then montin yeah so the alternative to perceptional learning would be like to destroy the ball or hide the ball or stop the motion of the ball or ignore the ball like there's all these other ways that would be neither perception nor learning I just want to add that yes um what I he about is this conceptual changes and I was thinking about uh what is the exact difference between an insight and recognition also just phenomenologically wise and I was really thinking about uh what this difference is related on on what level in the hierarchy uh this belief is changing so very deep beliefs up in the hierarchy those are more considered parad time shifts and those lower in the hierarchy if those beliefs are getting in there those are just recognitions and just becoming aware of the perceptual environment and I was thinking about how you guys reflect on this yeah because the uh if you go to the lower levels I mean an echo chamber in social media is a kind of Paradigm that it just operates at a totally different speed than Paradigm shifts in science but if you go to Echo chamber you actually have the if correct the higher level beliefs stay constant for the just only reinforcement so there's no inside transfer on higher level well they if you go back to Thomas Coon's definition of what a paradigm is it defines it defines the questions you're allowed to ask under that particular Paradigm and one of the one of the points he makes about paradigms is there's no last Paradigm you know it's it's like a fractal it's all the way down you never it's Rush dolls you never actually get to the end of the paradigms and and and you isolate yourself this is a new theory in AI into AI neighborhoods depending on the Paradigm you're operating with and that's got very different consequences in terms of like the world will continue to separate into different belief systems regardless of where AI goes it will be a continuous part of of of how intelligence evolves okay it seems that Paradigm shifts for me is different understanding but perhaps resolvement will become later just say I come to I've come to the same conclusion myself particularly relating to uh to you know religious uh events they are you know spiritual shifts major changes in perception they are the same thing as a messing with a belief at a high at a particularly deep level of one's predictive you can't help you have to rewrite the whole story all the way up right and you experience everything very different and I've had a few of those religious experiences which is why the question is kind of fundamental to me I've had a total re looking in a totally new Direction than I than I was looking before as a result of like a single very brief experience it's a phase transition from water to ice basically it's it's it's a radically different shaping of perceptions yeah and what this model does is it gives us a very clear way in which I would argue no no religion yet has in terms of how to bring that about how to trigger it well true but I'm back to the point is like there there is no such thing as the perfect Paradigm I mean even even believed that the old generation of scientists had to die off before you could actually get to a new paradigm because they it was too too hard to change it at the very end it's like the religious transition that St Augustine went through that kind of thing yeah so um Ali and then Andreas anyone else with their handr but yeah interesting yeah I just wanted to point out uh Michael Levan's recent uh article self-improving memory about um how learning or actually memory formation um occurs even uh in very basic uh complex agents such as uh molecular uh biological um uh I mean complex systems uh so basically uh the point here is that uh when when an agent learns about its environment it's not so much about uh the Fidelity or uh the exact modeling of of its environment but more uh the Salient parameters of the environment uh in which it tries to um preserve so uh the higher it goes in its hierarchical structure of its memory formation uh the more uh the more it needs to keep those Salient parameters or as John Veri would call uh the more it needs to uh realize the um to do the uh relevance realization that's the term uses uh so basically uh the thing about perception and learning and why uh they are closely interrelated uh I think depends uh in part uh on this process of memory formation uh I mean uh the higher I mean um the higher we go up onto uh on the latter of this um hierarchy iCal um generative modeling of of the environment uh the more it needs to retain uh the relevant and uh Salient parameters of the environment so on the very basic levels uh some of some of the uh perceived elements would not be needed uh for uh long-term perseverance of that agent through the environment but uh um as we know the longer the agent wants to persevere through the environment uh the more it needs to attend to those uh Salient and relevant uh parameters and one other thing I wanted to point out relating to this um interconnection between perception and action uh Alan Barto has a a famous code relating these two um parameters as saying uh which I think is beautiful uh he says perception is simulated action uh which will come to in chapter two why it can be also applied uh for active inference so uh yeah we can look at perception and action uh in a dual form from two uh complementary um viewpoints but at the end of the day they both somehow provide uh this the same formal uh structure looking whether we look at the environment or from the environment um to um from the agent to the environment or from the environment to the agent when you say perception is simulated action isn't that imagination not exactly you see uh when well it comes to as it plays on uh the elements of ecological Psych ology you see and the concept of affordances from the ecological psychology but of course affordances are used in a very different sense in active inference but um in brief when uh an agent sees an affordance an opportunity to act upon it doesn't perceive that object or that element of the environment just as a thing but it sees as it it SE the element as an opportunity to and how the agent can act upon that opportunity right how does that how is that different from imagination though uh well I'm not sure if I understood your question correctly but uh to my mind when we talk about imagination we don't necessarily use this kind of affordance thinking we don't map affordances onto the object of the environment as a very simple example when yeah if I see a gun in a movie in a drawer it's an old Hollywood thing is like you know somewhere later in the movie the gun's gonna get used right so affordance of gun is like the imagination starts looking ahead as like how is that gun gonna be used in the story in the narrative right yeah sure so as it may be just semantics imagination and you know and I'm trying to what I'm actually poking at is is there a real difference between imagination or just making predictions about what I could do H you know under under inference Theory that's Allis you see imagination plays um very significant role in shaping uh the prediction and uh even our generative model of the world but uh you see uh as a as a very simple example when we look at um um coffee M uh whether we want to uh I don't know uh hold on to that copy coffee M uh from uh from the rim or uh I don't know hold it in I mean um in an other way uh those two different perceptions would provide um a those two different ways of acting upon that coffee M would provide two different perceptions of that same object that's um it's it's it's a random thought but I'm building on what you're saying I'm not criticizing it it makes me think of like that's how topology works on mathematics what are the you know what are and so it's so there's a topological component of perception I suspect that yeah that will also return also just a general comment like crashing English words or other natural words into each other it's like continent scale discussion how is Imagination related to surprise or you know it how is a for it's just like they they have so it's such a broad scope that it's like it's absurd so it's great and it's like but it's the it's the it's the most symbolized coar grain token that we can share and reference so it's just like pointing it's like yeah you know um the North Pole is that way and so that is is like just a doorway and then there's the whole generative modeling process and then when you get down to it you have um analytical expressions and specific structured models that describe like well here's how I'm thinking about imagination in this setting this is the cognitive phenomena that I'm associating with imagination and that is let me can can I bring it back to something a little more data specific then because you I agree with you when you take the simple words you get the big Ideas when you take the big words you the little ideas I think that's the way I think of words H but are you familiar with the with the remembrance bump or the memory bump it's the point that memories in human lifespan are not evenly distributed but but they're actually concentrated between the years of 15 and 25 or 30 and the reason for that is because that's when you are going through the phase transition of the concept of self that you had that your parents imposed on you up until you became a teenager and then you're Reinventing yourself your adult sense of self and separating from the childhood sense of self so it speaks to a phase transition in a self-referential way to the concept of self not just the environment and I'm curious about as you know because I've got teenage grandkids as they're going through this biological process of basically creating their adult sense of self and separating from the parents idea of who they should be uh and there's a lot of data behind that and so on you can use it to build predictive models actually of like what that's why Hollywood does remakes of movies and stuff like that take the same story and update it with new generation and and stuff like that so when you introduce the self- referential component to the concept of self not just the environment how I'm going to be curious to see how active inference deals with that self-referential component yeah keep sketching it out and seeing what simpler generative model motifs you can bring in and you'll be well well well on the way towards that um Marvin and then Andreas with the hands rais and then anyone else and then that will be it I wanted to I've raised my hand sorry go ahead go ahead yeah um going back to the hierarchy uh for me um I'm interested in looking at that through the middle and I think what what I'd like to do with active inference is to say hey High going up there could could be a hierarchy going down but I think there's an essential gap between um the two minds let's say theing mind so thinking of an environment saying oh well what is the unconscious that's presenting this and is all these answers and then uh what is this mind that doesn't know but is able to ask questions and uh you know engage and investigate and so those would be like somewhere in the middle of the hierarchy and that way you can go further out into the world you can go off into abstractions but um you don't have to worry about them so much because the main focus is on how these very two very different ways of looking at things in terms of what we know and what we don't know how they match up that's what I'm interested to okay thank you um Marvin and then any final raised hands then that will be it okay uh quite basic I I'm I'm serous maybe just I know might be a deep topic but just basically like what's the relation what does active inference imply the predictive processing theory of the mind or predictive coding because what I really liked in the preface uh it says uh we're not simply trying to make sense of our s Sensations we are actively creating our sensorium so to me that sounded a lot like predictive uh processing or generative whatever you want to call it so so so is is predictive processing just derivation of active inference or an implication or is it yeah in parallel yeah great question um I I know that there's some question in one of the the chapters but um active predictive processing so considering the active prior meeting the observation like on the retina and including the capacity for the retina to move active predictive processing gets you it it it is active inference they basically are are are more isomorphic than not in live stream 43 um Maria does an awesome historical review like going back several thousand years to this concept of like the active predictive elements [Music] um they're more similar than not if you're talking about descending predictions ascending sensory information and then the descending predictions not just being well what will I but what will I do you basically have the scheme for octave inference and then there's some slight differences just um like more like historically culturally with like well predictive processing is often discussed in terms of a hierarchical model but active influence models they don't have to be hierarchical models they could just be single layer but that but predictive processing arising from like visual Mamon Neuroscience so then that's why that architecture became kind of generalized from visual perception versus just being in its most generic system independent form thank you thank you yeah okay Martin and then that will be it yes is a small question I think a small question uh to state but I think a very interesting question to think about and that is the relation between how to view willpower through the sense through the lens of active inference and my only gut feeling is that it has if there is certain way to look at it interpret it it would be in how you are able to resist uh prediction error like how a how much you are able to um have like this tension in the model of expected things and not the expected prediction uh percent I mean but that's a very interesting question I I'd like to yeah think about cool yeah and just just the predictive processing predictive coding basian brain that they they again they all are are closer than not but there's a few specific resources if you really want to split the hair okay last comment Andrew and then that will it yeah so on the will I think that's very much where I'm headed is I believe there's three Minds there's one that has these answers there's one that has these questions one that investigates and so it's the investigating mind that kind of pairs them up and sees oh uh do I want to lean on perception updating beliefs or do I want to lean on action updating the environment if now you could say it's just a free energy principle but the theory maybe allows for something more that there's a third mind that kind of actively uh applies that to equate those two but also to decide I'm G to use one you know I'm gonna lean this way I'm G to lean that way so then that would be very willful that's what I'm interested model the the's book The will the world is Will and representation or the world is Will and idea is probably a an interesting mapping onto active inference Theory cool thank you all um hopefully this is a little taste um of uh lineer conversation and like I can see that you all have many Avenues this makes you explore and I really encourage you to write them out so that they're simultaneous and then contribute them into our Niche where they're much more digestible than trying to linearly convey these ideas if you have them pre-loaded and you're thinking about it and you're wondering about like do that Chain of Thought and then we'll have something to really digest deliberately over hours and something that people can add to um rather than just follow up on so hence the sort of um live but mainly I mean we're you know next several hundred hours how do we do that again remind me just really quickly the whole doc everything we looked at with a document is editable so you can ask and add questions like by adding a row that was shown um but you can add to the discourses the canvas that was next to the question on the row okay or or or you can just you know write it on your own blog or make a GitHub repo and share that link if if if you don't see the right Nook or cranny in our shared Niche just write it and then link it in okay so all right thank you see you all in um one or two weeks all right bye